

Date

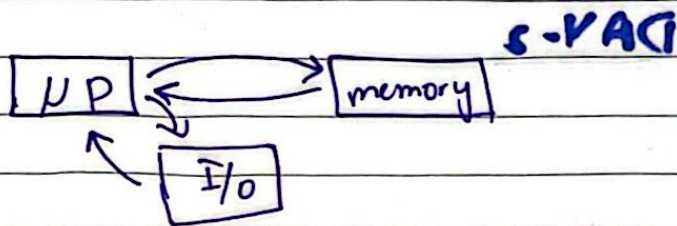
µP programming & interfacing

DAY-1

What does a µP do:

1. Fetch
2. Decode
3. Execute

→ Assembly language

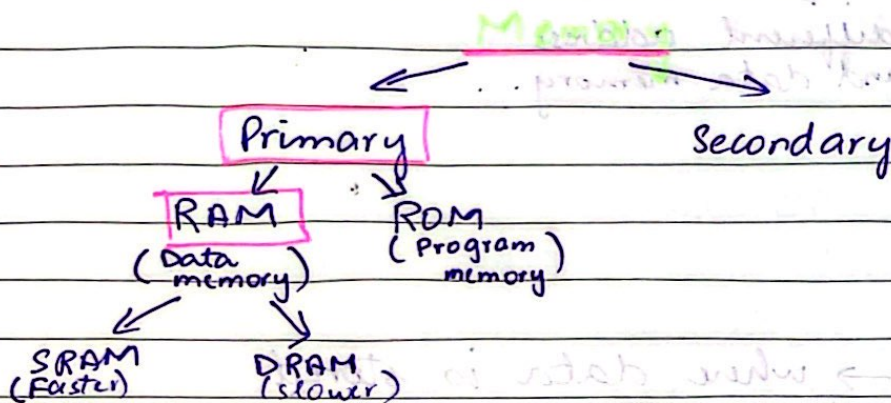


00 → ADD

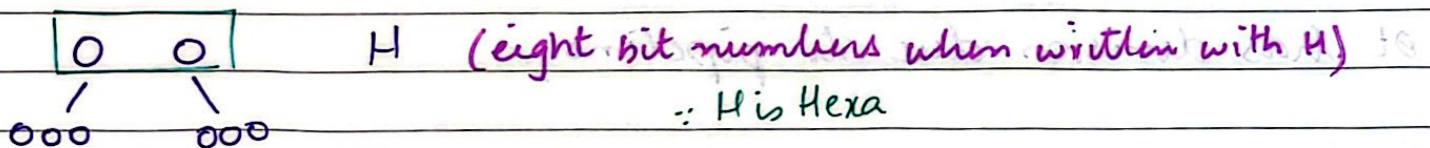
01 → SUB

10 → MUL

11 → DIV



We will use hexadecimal in assembly language.



Example:-

60 23 H

total 16

3	D	F	0	7
0	9	3	0	4
3	0	E	F	

Date

operation

ADD

Register

BL

SSH

operand

DAY-1

∴ there is no OP code for numbers

OP code will be converted
To 01 (Binary)

DAY-2

Architecture of 8086 :-

princeton
Von-Neumann

↓
Memory
RAM
ROM

↓ Harward

↓
different address
and data memory

same address
and data bus

- Memory interphase → where data is stored
- BIU → Bus interphase Unit
- EU → Execution Unit

It has division due to pipeline

Pipelining :-

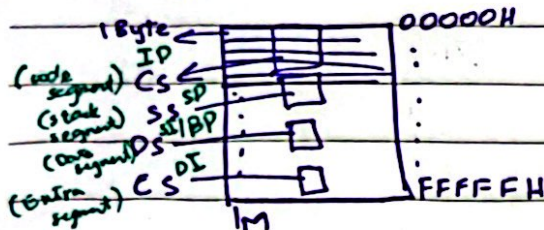
F	D	E	F	D	E
	F	D	E	F	D
		F	D	E	F

Advantage :-

- When one input is being executed, the next one is ready to be executed

Date

If it has 20 bit address bus, it can have 2^{20} bytes of memory 1MB.



∴ divides memory into 4 bytes parts

Fetch:-

starting address $\times 10 + \text{offset}$ (physical address)

- IP is instruction point which tells the offset.
- OP code is a series of ^{0s & 1s} ~~processors~~ that is understandable by computer.
- Instructions are converted into OP code.
- BIU has segment registers - carry starting address.
- IU has general purpose registers.
 - AX, BX, CX, DX = 2 byte = 16 bits
 - AH, AL, BH, BL, CH, CL, DH, DL = 1 byte = 8 bits
- Flags are indications that indicate particular condition.
- Operand is a number that is different than OP code. It is a temporary register.

Example:-

starting address = 5D27H

offset = 2C30H

Find the physical address

$$= (5D27 \times 10) + 2C30H$$

Date

$5D270H + 2C30H$

$5D270H$

$2C30H$

$5FEA0$

physical address = $5FEA0$